



ROOT CAUSE UNLOCKED · CONDITION KIT

The Hormone Root Cause *Kit*

Which tests actually answer the question, which ones do not, and exactly what to ask your doctor for

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The Hormone Root Cause Kit

Which tests actually answer the question, which ones do not, and exactly what to ask your doctor for

BEFORE YOU START

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Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Nothing in this kit is a reason to stop or change a prescription. Not birth control, not testosterone replacement, not metformin, not thyroid medication. If you want to change any of those, that is a conversation with the prescriber who started them.

This is not fertility treatment. If you are trying to conceive and it is not happening, that is a reproductive endocrinologist, and time matters. Do not spend a year on supplements first.

See a doctor promptly for: a breast lump, unexplained vaginal bleeding after menopause or between periods, bleeding after intercourse, sudden severe pelvic pain, visual changes with headaches, unexplained weight loss, or new severe fatigue with dizziness and darkening of the skin. That last combination can indicate adrenal insufficiency, which is a genuine and serious diagnosis.

The one idea this whole kit rests on

Key 1 of *Unlock Your Hormones* says it: **hormones are messengers, not the root cause.**

A hormone panel tells you what the messengers are doing. It rarely tells you why. Testosterone that is low because of poor sleep, excess weight, and untreated apnea reads the same on paper as testosterone that is low for a pituitary reason, and those need completely different responses. Estrogen that looks unbalanced because of insulin resistance is not the same problem as estrogen that looks unbalanced for a structural reason.

So the useful question is never “what is my hormone level.” It is “what is making it do that.”

The eight keys, and what tests each one

Key	What it means	How you find it
1. Messengers	Hormones report on the system, they are not the system	The framing for everything below
2. Insulin	Insulin resistance is the engine behind most PCOS	Fasting insulin, glucose, HbA1c. The highest-yield panel here
3. HPA axis	Chronic stress and its effect on sex hormones	Contested. See the honest section below
4. Estrogen metabolism	Clearance, the gut, and the estrobolome	Metabolite panels, read as routes rather than targets
5. Androgens	Not just a male concern	Total and free testosterone, SHBG, DHEA-S
6. Low testosterone in men	What is actually driving it under fifty	Morning total testosterone, repeated , then a cause hunt
7. Thyroid	The cross-referenced driver	TSH, free T4, free T3, antibodies
8. Testing honestly	Serum versus dried urine, and what each is good for	The section below is the one to read first

Key 8: serum, salivary, and urine, stated honestly

I am putting this first because it decides how much money you spend on everything else, and because the honest answer is more interesting than either side usually admits.

Serum testing is what most clinical decisions are made on, what most reference ranges are built from, and what your physician will act on. It remains the default.

Dried urine panels are widely used in functional medicine, and the common criticism of them, that they are simply unvalidated, is not accurate. Here is what the published record actually shows.

What the assay validation shows, which is real

A method study compared dried urine analyzed by gas chromatography with tandem mass spectrometry against serum, and against 24-hour collection. It reported good reliability between the dried urine and serum assays, urine metabolite concentrations following the same pattern as serum across the menstrual cycle in a subset of women, and intraclass correlation coefficients above 0.95 comparing the four-spot method with 24-hour collection and dried with liquid urine, which is excellent agreement. Its conclusion was that for estradiol and progesterone the dried urine assay is a good surrogate for serum testing (PMID 31384769).

A second study extended this across a wider panel, finding dried versus liquid urine agreement above 0.90 for reproductive hormones and 0.75 to 0.99 for organic acids, and four-spot versus 24-hour agreement above 0.9 for 14 of 17 metabolites with no systematic differences between collection methods (PMID 33722278).

A third study showed that dose-proportional changes in estrogen exposure from transdermal estradiol gel could be seen in the urinary estrogen profile, and distinguished the profiles of premenopausal women, postmenopausal women on estrogen therapy, and postmenopausal women on none (PMID 35483542).

So the assay measures what it claims to measure, reliably, and a four-spot dried collection can stand in for a 24-hour jug. That is genuine analytical validation and it deserves to be stated plainly.

What that does not establish, which is the part that matters for you

Analytical validation and clinical validation are different things.

Analytical validation asks whether the assay accurately and reproducibly measures the molecule. The answer above is yes.

Clinical validation asks whether making treatment decisions from those results leads to better outcomes than not doing so. **None of those studies answer that question**, and I could not find studies that do. That is not a criticism of the assay. It is a statement about what a measurement can justify on its own.

Two more things you should hear from me rather than discover later.

Who did the research. All three studies were conducted by the laboratory that sells the test, and the papers disclose it: the founder and owner is an author on each. Two include Frank Stanczyk, an independent steroid assay researcher whose involvement is meaningful, and one notes that the statistics were calculated and interpreted by an independent agent. Those disclosures are to their credit. It also remains true that company-run validation of a company's own product is weaker than independent replication, and independent replication is what I could not find.

How the reference ranges are built, which is the real caveat. This is not my inference. The manufacturer's own interpretive guide states it. The ranges come from their internal database with their own exclusions applied, and the guide says the ranges are deliberately **set more tightly than classic diagnostic ranges so as to be optimally used in functional medicine practice**, giving the example that the upper limit for testosterone is set at the 80th percentile in women and the 90th in men.

Read that carefully, because it is the crux. A classic diagnostic range is built to separate disease from health. A percentile cutoff chosen for clinical utility is built to flag people for attention. **Being outside such a range means you are in a minority of the tested population. It does not mean you have a disorder.** The manufacturer is transparent about this, and it is the single most useful thing to understand before you read your own report.

What I would actually do with one

- **If a decision matters, anchor it on serum**, particularly anything involving a diagnosis or a prescription.
- **Use dried urine for what the validation actually supports**: tracking a pattern over time, monitoring estrogen exposure on therapy, and seeing metabolite routes that serum does not show.
- **Treat an out-of-range metabolite as a prompt to look closer, not as a diagnosis**, for the reference-range reason above.
- **Do not repeat it often**. It is most informative as a baseline and a follow-up, not as a monthly scorecard.

The one to still be careful about: cortisol panels sold to diagnose “adrenal fatigue”

This is a separate claim from everything above, and the validation does not transfer to it.

The assay can measure cortisol and its metabolites accurately. That does not make **adrenal fatigue**, meaning stress-exhausted adrenals that under-produce, a recognized clinical diagnosis. It is not one. A pattern on a cortisol curve is not a disease, and measuring something accurately does not validate the interpretation sold on top of it.

Adrenal insufficiency is a real and serious diagnosis, established with morning cortisol and specialist testing rather than a wellness panel. If you have severe fatigue with dizziness, low blood pressure, salt craving, and darkening skin, that is a physician now, not a kit.

Key 2: insulin, the engine behind most PCOS

If you have PCOS, this is the highest-yield section in this guide.

What to ask for. Fasting insulin, fasting glucose, and HbA1c together. Fasting glucose alone will look normal for years while insulin climbs to keep it there, which is exactly the window in which something can be done.

Why it matters more than the sex hormones. The international guideline framing puts lifestyle intervention as first-line therapy in PCOS, and a modest weight loss of 5 to 10% can significantly reduce reproductive, metabolic, and psychological symptoms, while the same source is candid that achieving it remains a challenge for patients and clinicians alike (PMID 32814159).

Read both halves of that. The target is smaller than people assume, and the difficulty is real and acknowledged, not a failure of willpower.

PCOS diagnosis, briefly

PCOS is diagnosed on a combination of irregular ovulation, clinical or biochemical androgen excess, and ovarian appearance on ultrasound, with other causes excluded. **You do not need all three.** Two of three, with other causes ruled out, is the standard framework, and the exclusions matter: thyroid disease, elevated prolactin, and non-classical congenital adrenal hyperplasia can all imitate it.

What to ask for alongside the insulin panel: total and free testosterone, SHBG, DHEA-S, LH and FSH, prolactin, TSH with free T4, and 17-hydroxyprogesterone.

Key 6: low testosterone in men, and the diagnostic discipline that is usually skipped

How common it is. Prevalence estimates suggest 6 to 12% of men aged 40 to 49 have biochemically low testosterone, commonly associated with obesity, metabolic syndrome, and sleep apnea (PMID 41127744).

Read that association list, because it is the whole point of this key. In men under fifty, low testosterone is far more often a consequence of something than a primary gland problem. Excess weight, poor sleep, untreated sleep apnea, and metabolic syndrome all suppress it, and all of them are addressable.

The diagnostic discipline. The Endocrine Society clinical practice guideline on testosterone therapy in men with hypogonadism sets out how diagnosis and monitoring should be done, including monitoring that evaluates symptoms, adverse effects and compliance, measures serum testosterone and haematocrit, and evaluates prostate cancer risk in the first year of therapy (PMID 29562364).

The two rules most often broken:

1. **Draw it in the morning.** Testosterone has a daily rhythm and an afternoon sample can be misleadingly low.
2. **Confirm it on a second, separate morning sample** before anyone acts on it. A single low reading is not a diagnosis.

And if it is genuinely low, find out why before deciding what to do. That means LH and FSH to separate a testicular cause from a pituitary one, prolactin, ferritin and iron studies, and a sleep apnea assessment. Starting replacement without that cause hunt can bury a finding that mattered.

If you are already on testosterone replacement, nothing in this kit is a reason to stop it. That is a prescriber conversation, and stopping abruptly has consequences.

Key 3: the HPA axis, without the marketing

Chronic stress genuinely interacts with sex hormones. That is real physiology, and it is Key 3 of the ebook.

What does not follow from it is that a saliva panel can measure your “adrenal fatigue” and a supplement can fix it. Covered above in Key 8.

What is worth doing instead. Treat load as a clinical variable: sleep, work hours, caregiving, training volume, and undereating are all inputs, and all of them are more tractable than a cortisol curve. If there is genuine clinical suspicion of adrenal insufficiency, that is a morning cortisol and a physician, not a wellness panel.

Keys 4, 5 and 7: the rest of the panel

Estrogen metabolism and the estrobolome (Key 4). The gut participates in estrogen recirculation, which is real biology and is the rationale for the fiber emphasis in the diet guide. On testing: the metabolite measurements themselves are analytically sound, as covered in Key 8, and what has not been established is that steering treatment by metabolite ratios changes outcomes. So read the ratios as information about routes rather than as targets to correct. Fiber, bowel regularity, and alcohol moderation are defensible on their own terms without needing a ratio to justify them.

Androgens in women (Key 5). Total and free testosterone, SHBG, and DHEA-S. SHBG matters more than people expect, because it determines how much of your testosterone is actually free and active, and it falls with insulin resistance. A normal total testosterone with a low SHBG can mean a high free testosterone, which is the pattern that explains symptoms.

Thyroid (Key 7). TSH with free T4 and free T3, plus TPO and thyroglobulin antibodies if never checked. Thyroid disease imitates and worsens most of what is in this kit. As in the other kits, I have deliberately not published functional target ranges for thyroid markers, because the right target depends on age, pregnancy, and existing replacement.

Two markers in that table that get read wrong

Homocysteine, read correctly. Homocysteine is a methylation-cycle marker: it rises when B12, folate, or B6 run short, and with thyroid or kidney trouble. Do not let anyone wave off an elevated result as trivia. In my read this is one of the most consequential numbers in the panel, because it tells you the methylation machinery is running short somewhere findable. The way to act on it is not to chase the number with blind supplementation. Find which input is short, B12, folate, B6, thyroid, or kidney function, correct that, and let homocysteine confirm the fix on retest. My functional target of under 7.0 in the table below reflects how seriously I take it.

hs-CRP, read for what it is. It is a nonspecific screening marker. An elevated result tells you an inflammatory load exists somewhere. It does not tell you where it is coming from, and it never confirms that your hormones are the source. It is worth having because a load you did not know about changes what you look for next, not because the number itself points anywhere.

Functional ranges for the metabolic markers

Clinical targets I use at Root Health, not validated diagnostic thresholds. Sex hormone reference ranges are deliberately not listed here, because they depend on sex, age, cycle day, and menopausal status, and a single table would mislead more than it helps. Your lab's ranges and your clinician's reading are what count for those.

Marker	Units	Functional target	Approximate standard range
Glucose, fasting	mg/dL	75 to 86	70 to 99
HbA1c	%	4.6 to 5.3	4.0 to 5.6
hs-CRP	mg/L	under 0.5	under 1.0
Ferritin	ng/mL	50 to 150	25 to 300
Vitamin D (25-OH)	ng/mL	40 to 80	20 to 100
Vitamin B12	pg/mL	500 to 900	200 to 1100
Homocysteine	umol/L	under 7.0	under 10.0
Magnesium (RBC)	mg/dL	5.2 to 6.8	4.0 to 7.2
Zinc (plasma)	ug/dL	70 to 110	60 to 120
Albumin	g/dL	4.2 to 4.8	3.5 to 5.0

Reading your results as a pattern

Pattern one: PCOS with insulin resistance. Irregular cycles, androgen excess, fasting insulin creeping up, SHBG low. This is the most common and most treatable pattern in the kit, and the diet and supplement guides are built around it.

Pattern two: low testosterone with a cause attached. A man under fifty with low morning testosterone, excess central weight, and snoring. The testosterone is downstream. Treat the apnea and the weight and re-measure before deciding about replacement.

Pattern three: normal panel, real symptoms. Common. It moves the work to sleep, load, blood sugar stability, and thyroid, and it is not a dismissal.

Pattern four: something that needs a specialist. Very high testosterone or DHEA-S, high prolactin, visual changes, or postmenopausal bleeding. Those go to a physician, not to a protocol.

The conversation with your doctor

A request list you can bring

If PCOS is the question: fasting insulin, fasting glucose, HbA1c, total and free testosterone, SHBG, DHEA-S, LH, FSH, prolactin, TSH with free T4, 17-hydroxyprogesterone.

If low testosterone is the question: morning total testosterone, repeated on a second morning, plus free testosterone, SHBG, LH, FSH, prolactin, ferritin and iron studies, HbA1c, and a sleep apnea assessment.

For everyone: 25-hydroxy vitamin D, B12, hs-CRP, and a CBC.

The sentence that helps

“My fasting glucose is normal. Could we add a fasting insulin, because I understand glucose stays normal for a long time while insulin is compensating?” That is specific, accurate, and it is the test most often left off.

What these labs cannot tell you

- **Why your hormone is doing what it is doing.** That is Key 1, and it is the whole point.
- **Whether you have adrenal fatigue.** Not a recognized diagnosis.
- **Whether a urine metabolite pattern should change your treatment.** The measurement is sound. The outcome evidence for steering treatment by it is what is missing, and Key 8 explains the difference.
- **Whether you can conceive.** That is a fertility evaluation, and it is time-sensitive.

When to seek clinical review

Promptly: breast lump, postmenopausal bleeding, bleeding between periods or after intercourse, sudden severe pelvic pain, visual changes with headache, unexplained weight loss, or severe fatigue with dizziness, low blood pressure, salt craving and skin darkening.

Soon: no periods for three months outside pregnancy, rapidly worsening hirsutism or male-pattern hair loss, galactorrhoea, or twelve months of trying to conceive without success, six months if you are over 35.

Your next step

The companion guides do the rest: **The Hormone Diet Guide** covers insulin resistance and the fiber and estrobolome work, and **The Hormone Supplement Guide** covers inositol, NAC, and what the evidence does and does not support.

Book a free Root Discovery Session: rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=hormones-labs

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

Get in touch: rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=hormones-labs

If you would rather start on your own, track your cycle and symptoms together. In this category the pattern across three cycles is worth more than any single panel:

Free symptom tracker: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=hormones-labs

References

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1. Lie Fong S, Douma A, Verhaeghe J. Implementing the international evidence-based guideline of assessment and management of polycystic ovary syndrome (PCOS): how to achieve weight loss in overweight and obese women with PCOS? *J Gynecol Obstet Hum Reprod*. 2021;50(6):101894. PMID 32814159
2. Canal de Velasco LM, Gonzalez Flores JE. Testosterone Therapy in Men in Their 40s: A Narrative Review of Indications, Outcomes, and Mid-Term Safety. *Cureus*. 2025;17(9):e92778. PMID 41127744
3. Bhasin S, Brito JP, Cunningham GR, et al. Testosterone Therapy in Men With Hypogonadism: An Endocrine Society Clinical Practice Guideline. *J Clin Endocrinol Metab*. 2018;103(5):1715-1744. PMID 29562364
4. Newman M, Pratt SM, Curran DA, Stanczyk FZ. Evaluating urinary estrogen and progesterone metabolites using dried filter paper samples and gas chromatography with tandem mass spectrometry (GC-MS/MS). *BMC Chem*. 2019;13(1):20. PMID 31384769
5. Newman M, Curran DA. Reliability of a dried urine test for comprehensive assessment of urine hormones and metabolites. *BMC Chem*. 2021;15(1):18. PMID 33722278
6. Newman MS, Curran DA, Mayfield BP, Saltiel D, Stanczyk FZ. Assessment of estrogen exposure from transdermal estradiol gel therapy with a dried urine assay. *Steroids*. 2022;184:109038. PMID 35483542

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Educational content only. Not medical advice. Not a diagnosis. Not fertility treatment. Not a reason to change any prescription.

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Nothing here is a reason to stop or change a prescription, including birth control, testosterone replacement, metformin, or thyroid medication.

This is not fertility treatment. If you are trying to conceive without success, that is a reproductive endocrinologist, at twelve months of trying or six months if you are over 35.

If you are pregnant, nursing, underweight, or have a history of disordered eating, talk with a clinician before changing how you eat. **This guide discusses weight, and it does so carefully, for reasons in the next section.**

See a doctor promptly for a breast lump, postmenopausal bleeding, bleeding between periods, sudden severe pelvic pain, or visual changes with headache.

Before anything else, about weight

The guideline literature puts lifestyle intervention as first-line therapy in PCOS, and a **weight loss of 5 to 10% can significantly reduce reproductive, metabolic, and psychological symptoms**, while the same source states plainly that achieving it remains a challenge for both patients and healthcare providers (PMID 32814159).

Two things follow, and I want both on the record.

The target is smaller than most people are told. Not a transformation. Five to ten percent.

And the difficulty is real and documented, not a character flaw. The exercise position statement for PCOS specifically asks clinicians to account for the significant psychological burden, including **weight stigma**, and the high prevalence of comorbidities, when delivering interventions (PMID 38960811).

If you have spent a decade being told to lose weight by people who did not help you do it, that position statement is on your side. This guide tries to be the help rather than the instruction.

And if weight is not your issue, which is true for a meaningful share of people with PCOS, skip this and go to Part Two. Lean PCOS is real and the insulin work still applies.

Where food sits among the eight keys

Unlock Your Hormones works through eight root causes. Food is a direct lever on Key 2, insulin resistance, which the ebook calls the engine behind most PCOS, and a real contributor to Key 4, estrogen metabolism and the estrobolome. It supports Key 6 in men, where weight and metabolic health drive much of what looks like a testosterone problem.

It is not a lever on thyroid autoimmunity, and it is not a treatment for infertility. The lab guide maps all eight keys.

Part one: eating for insulin, which is the engine

If you take one thing from this guide, take this part.

The goal is not a low-carbohydrate diet. It is a diet that does not spike insulin repeatedly across the day. Those are different, and the second is much easier to live with.

The five changes that do most of the work:

1. **Protein at every meal, starting with breakfast.** This single change flattens the rest of the day more reliably than anything else here.
2. **Never eat carbohydrate naked.** Pair it with protein, fat, or fiber. Toast alone and toast with eggs behave completely differently.
3. **Fiber at every meal.** Vegetables, legumes, whole grains, and fruit with the skin on. Fiber does double duty here, and Part Three explains the second job.
4. **Cut liquid sugar first.** Sweetened drinks, juice, and the coffee order that is functionally a dessert. This is usually the largest single input and the easiest to remove.
5. **Walk after meals.** Ten to fifteen minutes. It blunts the post-meal glucose rise and it is the least demanding exercise intervention that exists.

What to expect and when. Energy and afternoon crashes often shift within two weeks. Cycle changes take cycles, usually three to six months. Judge accordingly.

Sourcing note. The five practices above are standard clinical nutrition for insulin resistance rather than findings from a PCOS-specific trial, and I have not attached citations to them for that reason. The guideline-level claim they serve, that lifestyle is first-line and 5 to 10% weight loss meaningfully reduces symptoms, is cited above (PMID 32814159).

Part two: movement, because it is a hormonal intervention here

Exercise in PCOS has its own position statement (PMID 38960811), which is unusual and tells you the evidence base is taken seriously.

What matters most, practically:

- **Resistance training two or three times a week.** Muscle is where glucose goes. This is the highest-value form of exercise for insulin resistance and it is chronically under-prescribed to women.
- **Walking daily**, especially after meals.
- **Something you will still be doing in a year.** Adherence beats optimality, and the position statement's point about psychological burden applies here too.

For men with low testosterone, the same logic runs harder. A scoping review of nonpharmacological interventions concludes that proper exercise, sleep, and diet are indisputable approaches on general health grounds, while nutraceuticals rest on weaker evidence and need personalizing (PMID 35810031). Weight, sleep, and apnea are the levers, and they are the ones a capsule cannot reach.

Part three: fiber, the gut, and estrogen

This is Key 4, and it is the part where diet does something specific rather than general.

The mechanism, plainly. Estrogen is conjugated in the liver and sent into the bowel for elimination. Gut bacteria can deconjugate some of it, allowing reabsorption. That recirculation loop is the estrobolome, and the two things that most influence how much gets reabsorbed are transit time and fiber.

What follows:

- **Fiber, aiming for 25 to 30 g daily**, built up gradually so you do not simply bloat.
- **A daily bowel movement.** Unglamorous, and directly relevant to the loop above.
- **Cruciferous vegetables** several times a week. Broccoli, cabbage, cauliflower, Brussels sprouts.
- **Alcohol moderation**, because the liver is doing the conjugation step.

What I am not going to tell you. That you need a urine estrogen metabolite panel to guide any of this, or that a “detox” supplement corrects the ratios on one. The lab guide covers that testing fairly: the measurements are analytically sound, and what has not been shown is that steering treatment by the ratios changes outcomes. Fiber, regularity, and alcohol moderation are defensible on their own terms and do not need a ratio to justify them.

Part four: a week that actually works

Breakfast, the highest-leverage meal. Eggs with vegetables. Greek yogurt with berries and seeds. Leftover dinner protein. Not cereal, not toast alone, not juice.

Lunch. A protein, a large volume of vegetables, a fiber-rich carbohydrate. Lentil soup with bread. Chicken and bean salad. Leftovers from last night.

Dinner. Protein, two vegetables, a starch you tolerate. Cook enough to cover tomorrow's lunch.

If you snack, pair it. Apple with nut butter, not apple alone.

The batch habit. Cook two proteins and roast a tray of vegetables on one day. It removes the decision at the point when you have the least capacity to make it.

What I deliberately left out, and why

- **Very low carbohydrate and ketogenic diets as the default.** They can help insulin resistance and they are demanding, and the guideline framing is lifestyle intervention with a 5 to 10% weight loss target rather than a specific macronutrient extreme (PMID 32814159). If you want to try one, it is a conversation, not a starting point.
- **Dairy and gluten elimination for everyone with PCOS.** Widely recommended, and I did not find evidence supporting blanket removal for this condition. Test if you have a reason.
- **Estrogen “detox” diets built on metabolite panels.** See Part Three.
- **Seed cycling.** Popular and I could not find trial evidence for it. Eating seeds is fine. The cycling schedule is the part without support.
- **Extreme calorie restriction.** It suppresses the reproductive axis, which is the opposite of the goal, and it makes the 5 to 10% harder to sustain.
- **Anything framed as a fertility diet.** If conception is the goal, the appointment is the intervention.

Telling you what to skip is part of what you paid for.

Running a fair experiment

Cycles 1 and 2, baseline. Change nothing. Track cycle length, symptoms, energy, and the things that bother you most.

Cycles 3 to 6, the insulin work. Protein at breakfast, pair the carbohydrate, fiber up, liquid sugar out, walk after meals.

Cycle 7, judge it. Compare cycles, not days. Cycle regularity is the outcome that matters most and it is slow.

Realistic expectations

What is realistic. Steadier energy in two weeks. More regular cycles over three to six months. Gradual improvement in metabolic markers. Slow improvement in hirsutism, because hair growth is slow.

What is not realistic. Diet curing PCOS, which is a long-term condition that gets managed. Diet fixing testosterone that is low because of untreated sleep apnea. Any of this substituting for a fertility evaluation.

When to seek professional care

Promptly: breast lump, postmenopausal bleeding, bleeding between periods or after intercourse, sudden severe pelvic pain, visual changes with headache.

Soon: no periods for three months outside pregnancy, rapidly worsening hirsutism or hair loss, milk discharge from the breast, twelve months trying to conceive (six if over 35), or a diet that has narrowed below roughly twenty foods.

Working with Root Health

This guide is written to be safe and useful for a reader I have never met. A personalized plan starts from your labs, your cycles, your medication list, and whether conception is a goal.

Book a free Root Discovery Session: rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=hormones-diet

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Read the companion documents alongside this one. **Understanding Your Hormone Labs** covers the fasting insulin most often left off the panel, and **The Hormone Supplement Guide** covers inositol and NAC with their honest evidence grades.

References

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2. Sabag A, Patten RK, Moreno-Asso A, et al. Exercise in the management of polycystic ovary syndrome: A position statement from Exercise and Sports Science Australia. *J Sci Med Sport*. 2024;27(10):668-677. PMID 38960811
3. Santos HO, Cadegiani FA, Forbes SC. Nonpharmacological Interventions for the Management of Testosterone and Sperm Parameters: A Scoping Review. *Clin Ther*. 2022;44(8):1129-1149. PMID 35810031

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This is not fertility treatment. If you are trying to conceive and it is not happening, see a reproductive endocrinologist. Twelve months of trying, or six if you are over 35. Time matters and supplements are not a substitute for that appointment.

If you are pregnant, trying to conceive, or nursing, treat every entry here as ask-first rather than try-it.

See a doctor promptly for a breast lump, postmenopausal bleeding, bleeding between periods, sudden severe pelvic pain, visual changes with headache, or severe fatigue with dizziness, salt craving and skin darkening.

Supplements in the United States are regulated as food, not as drugs. Quality varies. Look for independent third-party testing.

Why nutrient evidence looks thinner than it is

A large outcome trial costs tens to hundreds of millions of dollars, and that money only exists when a patent can pay it back. You cannot patent magnesium, an herb, or a food. So the evidence hierarchy this guide grades against is not neutral: it systematically over-documents patentable drugs and under-documents nearly everything else. That is measurable, not rhetorical. A Cochrane methodology review found that manufacturer sponsorship of drug and device studies produces more favourable results and conclusions than independent funding, through a bias that standard quality checks do not catch (PMID 28207928).

Two honest consequences follow, and they pull in opposite directions.

First, “no evidence” has three different meanings, and this guide tries to always tell you which one applies. *Tested and failed:* a real trial ran and the answer was no. *Never properly tested:* nobody could afford the trial, so absence of evidence is not evidence of absence. *Tested small and won:* real but modest trials, usually measuring markers rather than outcomes, because markers are what a small budget can afford.

Second, the asymmetry earns nutrients humility, not automatic credit. When isolated nutrients have received large publicly funded trials, the results have been mixed: some genuine wins, some nulls, and some harms. Small nutrient trials also carry their own publication bias toward positive results, just as drug trials tilt toward their sponsor. So this guide applies one rule in both directions: name the funding, name the evidence tier, and let you see both.

What outranks everything in this guide

For PCOS, lifestyle is first-line therapy, and the target is smaller than people think. A weight loss of 5 to 10% can significantly reduce reproductive, metabolic, and psychological symptoms and is recommended as first-line management, and the same source is candid that achieving it remains a real challenge for patients and clinicians (PMID 32814159).

Exercise has its own position statement, from Exercise and Sports Science Australia, which also asks clinicians to account for the significant psychological burden in PCOS, including weight stigma, and the high prevalence of comorbidities, when delivering exercise interventions (PMID 38960811).

I am quoting that second half deliberately. If you have been handed “just lose weight” as advice for a decade, the guideline literature is on your side about how inadequate that is as a plan.

For low testosterone in men under fifty, the cause is usually addressable. Prevalence estimates put biochemically low testosterone at 6 to 12% of men aged 40 to 49, commonly associated with obesity, metabolic syndrome, and sleep apnea (PMID 41127744). A scoping review of nonpharmacological interventions concludes that **proper physical exercise, sleep, and diet are indisputable approaches** because of their general health benefits, while nutraceuticals rest on weaker evidence and should be personalized by a physician or dietitian (PMID 35810031).

Sleep apnea in particular is worth naming: it suppresses testosterone, it is common in exactly this group, and no supplement substitutes for treating it.

Key 3 in practice: the stress axis, and why it is not a supplement problem

Key 3 of the ebook covers the HPA axis and sex hormones under chronic stress. It is real physiology and it is the key most often converted into a product it should not be.

What not to buy. Glandular and adaptogen protocols sold on the promise of fixing “adrenal fatigue.” That diagnosis is not recognized, and the lab guide covers why. Adrenal insufficiency, by contrast, is a real and serious diagnosis, and it is not found with a saliva kit; that one belongs with a physician.

On the testing itself. I do run four-point diurnal cortisol, salivary, in my practice. It is useful for the shape of the curve across a day, which is a different question from diagnosing a named adrenal condition, and it is worth separating those two things because the marketing in this space deliberately blurs them.

What is worth doing, and it is free. The exercise position statement in PCOS specifically asks clinicians to account for the significant psychological burden, including weight stigma, and the high prevalence of comorbidities when delivering interventions (PMID 38960811). That is guideline-level acknowledgement that the psychological load in this condition is part of the clinical picture rather than a side issue.

So treat load as a variable you manage rather than a panel you measure:

- **Five to ten minutes daily of slow breathing**, longer exhale than inhale. Consistency beats technique, and it costs nothing.
- **Protect sleep**, seven to nine hours on a fixed schedule. Short sleep worsens insulin resistance directly, which puts it upstream of Key 2.
- **Do not stack under-eating on top of hard training**. In this category that combination suppresses the reproductive axis, which is the opposite of the goal.
- **Name the stigma if it applies**. If you have spent years being handed weight advice instead of help, that is a load in itself, and it is worth saying out loud to a clinician who will listen.

Sourcing note: the breathing and sleep practices above are clinical reasoning rather than findings from a hormone-specific trial, and no citation is attached to them for that reason. They are included because they are free, carry no interaction risk, and act on a key the ebook identifies.

How to read this guide

Test first. The lab guide covers what. Fasting insulin is the one most often left off and it changes what you do here.

Give it three to six months. Cycle-related outcomes need cycles to measure, so a six-week trial answers nothing.

One change at a time.

Tier 1: inositol, for PCOS

This is the best-supported supplement in the whole kit, and it still needs its caveats stated.

What the research shows. An umbrella review synthesizing meta-analyses of randomized controlled trials assessed inositol in PCOS and found supporting evidence, while stating plainly that **outcomes based on low-quality evidence require cautious interpretation and should not solely guide clinical decisions**, and calling for larger, rigorous trials (PMID 41757236).

A systematic review and meta-analysis of combination therapy found that metformin plus inositol was associated with cycle regularization and reductions in hirsutism and the LH to FSH ratio compared with metformin alone, while noting that further studies are needed to clarify the true benefits (PMID 39331347).

A network meta-analysis comparing GLP-1 receptor agonists, metformin, and inositol on anthropometric and metabolic outcomes concluded that **no claims of overall superiority in PCOS management can be made** from that analysis, and called for long-term head-to-head trials across a broader range of outcomes (PMID 42490840). That paper carries a published correction, which is not a retraction.

And a useful nuance. A systematic review and meta-analysis found that **metabolic phenotype modifies the endocrine response** to inositol, meaning who you are metabolically may predict whether it helps your androgen picture. The authors describe this as a hypothesis-generating framework needing adequately powered, phenotype-stratified trials with clinically meaningful outcomes including standardized hirsutism and ovulatory measures (PMID 41947399).

How I read all that. Inositol is reasonable to try in PCOS, especially in the insulin-resistant phenotype, it is well tolerated, and the honest evidence grade is moderate at best with the umbrella review's own caution attached. It is not established as better than metformin, and the network meta-analysis says so.

Typical studied range. Trials most often use myo-inositol at around 4 g daily, frequently in a 40 to 1 ratio with D-chiro-inositol, taken in divided doses.

Cautions. Generally well tolerated, with mild gastrointestinal upset at higher doses. **If you take metformin, do not swap it for inositol on the strength of this guide.** The combination data above is about adding, not replacing, and that is a prescriber decision.

Tier 2: N-acetylcysteine, for PCOS

What the research shows. A systematic review and meta-analysis found NAC had a major effect on progesterone, endometrial thickness, and LH levels in women with PCOS, while finding **no significant differences in estradiol, SHBG, or FSH**, and concluding it may be a potential treatment option (PMID 39861414). A separate systematic review and meta-analysis found NAC supplementation decreased total testosterone and increased FSH, suggesting it might improve reproductive system function in PCOS (PMID 36597797).

How to read it. Two meta-analyses, both positive on some hormonal measures and explicitly null on others. Note what the outcomes are: these are biochemical and ultrasound measures rather than “you will feel better,” and the two reviews do not agree on FSH direction, which is worth knowing.

Typical studied range. Trials commonly use 1.8 g daily in divided doses.

Cautions. Nausea is the common complaint. NAC can affect blood clotting at higher doses, which matters with anticoagulants and before surgery. Ask first in asthma.

Tier 3: correct what is measurably low, and ask first about the rest

Vitamin D, B12, ferritin, and magnesium. Test, correct a real shortfall, retest. In this population ferritin matters particularly for women with heavy cycles, where the threshold argument goes further than most reference ranges suggest.

Zinc and magnesium for testosterone appear in every men’s supplement aimed at this. The honest grade comes from the scoping review above: nutraceuticals for testosterone rest on weaker evidence than exercise, sleep, and diet, and should be personalized rather than taken speculatively (PMID 35810031). Correct a measured deficiency. Do not expect a supplement to raise normal testosterone.

Ask your prescriber first about:

- **Anything while on birth control, TRT, metformin, or thyroid medication**, which is most readers of this guide.
- **Anything at all if you are trying to conceive, pregnant, or nursing.**
- **Berberine**, which has real metabolic activity and real drug interactions, and which behaves more like a medication than a vitamin.
- **DHEA**, which is a hormone rather than a nutrient, is not benign, and should not be self-started.

What I deliberately left out, and why

- **DHEA and pregnenolone bought over the counter.** These are hormones. Self-prescribing them without testing and monitoring is not the same category as taking magnesium, and the androgenic side effects in women are real.
- **“Adrenal support” glandulars and cortisol protocols.** Adrenal fatigue is not a recognized diagnosis, and the panels sold to diagnose it are covered in the lab guide.
- **Estrogen “detox” protocols built on urine metabolite ratios.** The measurements themselves are analytically sound, and the lab guide covers that evidence fairly. What has not been shown is that steering supplements by those ratios changes outcomes, so I am not going to sell a protocol aimed at correcting a ratio.
- **Herbal blends marketed to “balance hormones.”** That phrase does not describe a mechanism, and the blends typically contain several actives at doses below anything studied.
- **Anything sold as a fertility treatment.** If conception is the goal, the appointment is the intervention.
- **Testosterone boosters generally.** The evidence grade is weak (PMID 35810031), and in men under fifty the cause is usually weight, sleep, or apnea, which a capsule does not touch.

Telling you what to skip is part of what you paid for.

Running a fair trial

Months 1 and 2, baseline. Track cycles, symptoms, and the two or three things that actually bother you most. Get the labs from the lab guide.

Months 3 to 8, one intervention. For PCOS with insulin resistance, inositol alongside the diet and exercise work is the reasonable first move.

Month 9, judge it. Compare cycles rather than days. Cycle regularity, hirsutism, and how you feel are the outcomes that matter, and a repeat panel is secondary.

Realistic expectations

What is realistic. More regular cycles over three to six months. Gradual improvement in metabolic markers. Slow improvement in hirsutism, which responds over many months because hair growth is slow.

What is not realistic. A supplement resolving PCOS, which is a long-term condition managed rather than cured. A capsule raising testosterone that is low because of untreated apnea. Any of this replacing a fertility evaluation.

Timeline. Three to six months minimum, judged in cycles.

When to seek professional care

Promptly: breast lump, postmenopausal bleeding, bleeding between periods or after intercourse, sudden severe pelvic pain, visual changes with headache, or severe fatigue with dizziness, low blood pressure, salt craving and skin darkening.

Soon: no periods for three months outside pregnancy, rapidly worsening hirsutism or hair loss, milk discharge from the breast, or twelve months of trying to conceive, six if over 35.

Where to find quality versions of these

Nothing here names a brand, on purpose.

Look for third-party testing. NSF, USP Verified, or equivalent.

Match the form and ratio to the trials. Myo-inositol at around 4 g daily, often with D-chiro-inositol in a 40 to 1 ratio, is what the literature used.

Read the whole panel. “Hormone balance” blends frequently contain DHEA or botanicals with hormonal activity that are not on the front of the label.

If you would rather not evaluate all of that yourself, I keep a curated dispensary of professional-grade products at Root Health, screened for the standards above.

Root Health dispensary: us.fullscript.com/welcome/drseay

Disclosure: Root Health earns a portion of dispensary purchases. That revenue supports the practice, and it is also why this guide was written to be complete and usable on its own. Nothing here was included, excluded, ranked, or dosed based on what it earns, and the first-line interventions in this category, movement, sleep, and food, earn us nothing.

Working with Root Health

Everything here is general education, written to be safe for a reader I have never examined. A personalized plan starts from your labs, your cycle history, your medication list, and whether conception is a goal, because that last one changes the whole approach.

***Book a free Root Discovery Session: [rootcauseunlocked.com/discovery?](https://rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=hormones-supplement)
[utm_source=guide&utm_medium=ebook&utm_campaign=hormones-supplement](https://rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=hormones-supplement)***

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

***Get in touch: [rootcauseunlocked.com/contact?](https://rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=hormones-supplement)
[utm_source=guide&utm_medium=ebook&utm_campaign=hormones-supplement](https://rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=hormones-supplement)***

If you would rather start on your own, track first:

***Free symptom tracker: [rootcauseunlocked.com/tracker?](https://rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=hormones-supplement)
[utm_source=guide&utm_medium=ebook&utm_campaign=hormones-supplement](https://rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=hormones-supplement)***

Read the companion documents alongside this one. **Understanding Your Hormone Labs** covers what serum, saliva, and urine testing can and cannot tell you, and **The Hormone Diet Guide** covers the insulin work that sits underneath most of this.

References

All references were verified against live PubMed records on 2026-08-09, including a publication type and corrections check for retractions and errata. Verification detail is documented in [Hormones-Kit-Citation-Ledger.md](#).

1. Lie Fong S, Douma A, Verhaeghe J. Implementing the international evidence-based guideline of assessment and management of polycystic ovary syndrome (PCOS): how to achieve weight loss in overweight and obese women with PCOS? *J Gynecol Obstet Hum Reprod*. 2021;50(6):101894. PMID 32814159
2. Sabag A, Patten RK, Moreno-Asso A, et al. Exercise in the management of polycystic ovary syndrome: A position statement from Exercise and Sports Science Australia. *J Sci Med Sport*. 2024;27(10):668-677. PMID 38960811
3. Canal de Velasco LM, Gonzalez Flores JE. Testosterone Therapy in Men in Their 40s: A Narrative Review of Indications, Outcomes, and Mid-Term Safety. *Cureus*. 2025;17(9):e92778. PMID 41127744
4. Santos HO, Cadegiani FA, Forbes SC. Nonpharmacological Interventions for the Management of Testosterone and Sperm Parameters: A Scoping Review. *Clin Ther*. 2022;44(8):1129-1149. PMID 35810031
5. Duan M, Yang M, Li C, et al. Effects of inositol in women with polycystic ovary syndrome: an umbrella review of meta-analyses from randomized controlled trials. *Front Endocrinol (Lausanne)*. 2026;17:1741509. PMID 41757236
6. Kelly FA, de Oliveira Macena Lobo A, Cardoso JHCO, et al. Comparison of metformin with inositol versus metformin alone in women with polycystic ovary syndrome: a systematic review and meta-analysis of randomized controlled trials. *Endocrine*. 2025;87(2):389-399. PMID 39331347
7. Omarion A, Ayasa Y, Omarion Z, et al. Comparative analysis of glucagon-like peptide-1 receptor agonists, metformin, and inositol in improving anthropometric and metabolic outcomes in women with polycystic ovary syndrome: a network meta-analysis. *Front Endocrinol (Lausanne)*. 2026;17:1833904. PMID 42490840
8. Tienforti D, Puocci G, Venditti C, et al. Metabolic Phenotype Predicts Biochemical Response to Inositol Supplementation in Polycystic Ovary Syndrome: A Systematic Review and Meta-Analysis. *Clin Endocrinol (Oxf)*. 2026;105(2):237-244. PMID 41947399
9. Vina I, Vina JR, Carranza M, et al. Efficacy of N-Acetylcysteine in Polycystic Ovary Syndrome: Systematic Review and Meta-Analysis. *Nutrients*. 2025;17(2). PMID 39861414
10. Shahveghar Asl Z, Parastouei K, Eskandari E. The effects of N-acetylcysteine on ovulation and sex hormones profile in women with polycystic ovary syndrome: a systematic review and meta-analysis. *Br J Nutr*. 2023;130(2):202-210. PMID 36597797
11. Lundh A, Lexchin J, Mintzes B, et al. Industry sponsorship and research outcome. *Cochrane Database Syst Rev*. 2017;2(2):MR000033. PMID 28207928

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